BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The urinary system

The written notes for the urinary system lecture: the kidney and its layers, the sectioned kidney, the nephron, the JGA and filtration membrane, renal blood flow, the ureters, bladder, and urethra, and the urinary disorders. Module 5 competencies 133 to 140, with nine worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 133. Kidney gross anatomy and layers

In your course notes: Module 5 packet, Chapter 1, competency 133 (page 5).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Describe the location and tissue layers of the kidney and identify the hilum, capsule, adipose capsule, and renal fascia.

The urinary system is made of two kidneys, two ureters, the urinary bladder, and the urethra. The kidneys filter the blood and make urine. The rest of the system carries the urine away, stores it, and releases it.

Structure	What it is
Kidneys	The paired organs that filter the blood and form urine
Ureters	The paired tubes that carry urine from the kidneys to the bladder
Urinary bladder	The muscular sac that stores urine
Urethra	The tube that carries urine from the bladder to the outside
Roles of the kidney	Filtering wastes from the blood, balancing water, ions, and pH, and helping regulate blood pressure

Where the kidneys sit

- The kidneys are bean-shaped organs on the posterior abdominal wall, one on each side of the vertebral column, roughly between the T12 and L3 vertebrae. The lowest ribs partly cover their upper ends.
- The kidneys are retroperitoneal. This means they sit behind the parietal peritoneum, outside the peritoneal cavity, against the muscles of the back.
- The right kidney sits a little lower than the left kidney because the liver takes up the space above it.
- A suprarenal (adrenal) gland sits on top of each kidney like a cap. The adrenal gland is part of the endocrine system, but the same renal fascia wraps it together with the kidney.

The hilum

Each kidney has a rounded lateral border and a concave medial border, and the medial border faces the spine. The notch in the medial border is the renal hilum. The renal artery, renal vein, nerves, lymphatic vessels, and ureter all enter or leave the kidney here. Just inside the hilum is the

renal sinus, a space filled with fat that holds the renal pelvis, the calyces, and the branching vessels.

The three tissue layers, from the inside out

Three layers of tissue surround each kidney. From the kidney outward they are the renal capsule, the adipose capsule, and the renal fascia. Each layer has its own job.

Layer	Tissue	Job
Renal capsule (innermost)	A thin, tough layer of dense fibrous connective tissue on the kidney surface	Holds the kidney's shape and protects its surface
Adipose capsule (middle)	A layer of fat around the renal capsule, also called perirenal fat	Cushions the kidney and helps hold it in place
Renal fascia (outermost)	A sheet of dense connective tissue	Anchors the kidney and adrenal gland to the posterior abdominal wall and nearby structures

- Memory tip: go from the kidney outward: capsule, fat, fascia. The capsule sits on the kidney itself, the fat cushions it, and the fascia is farthest out because it ties the whole package to the body wall.
- The renal fascia is the outermost layer, not the capsule. The capsule is the smooth fibrous covering right on the kidney.
- The hilum is on the medial border, facing the spine. On a drawing, both hilums point inward, not out to the sides.

Competency 134. Internal kidney regions

In your course notes: Module 5 packet, Chapter 1, competency 134 (page 6).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Identify the cortex, medulla, renal columns, pyramids, papillae, parenchyma, calyces, and renal pelvis on a sectioned kidney.

When a kidney is cut in half from top to bottom (a frontal section), you see two kinds of tissue and a set of collecting spaces. The tissue that makes urine is called the parenchyma. The collecting spaces carry the finished urine to the ureter.

The working tissue

Renal cortex
: The outer layer, just under the renal capsule. It looks grainy because it holds the renal corpuscles and the coiled tubules of the nephrons.

Renal medulla
: The deeper region. It is made of cone-shaped renal pyramids, which look striped because tubules and collecting ducts run side by side through them.

Renal pyramid
: A striped cone in the medulla. Each pyramid has a wide base facing the cortex and a narrow tip pointing toward the renal sinus.

Renal papilla
: The narrow tip of each pyramid. Urine drains out of tiny openings on the papilla into a minor calyx.

Renal column
: Cortex tissue reaching inward between neighboring pyramids. One pyramid together with the cortex around it makes up one lobe of the kidney.

Parenchyma
: The cortex and the medulla together, the working tissue of the kidney. All of the nephrons are in the parenchyma.

The spaces that collect urine

Each papilla drains into a cup called a minor calyx. Several minor calyces join to form a major calyx, and the major calyces (usually two or three) join to form the renal pelvis. The renal pelvis is a funnel in the renal sinus that narrows into the ureter at the hilum.

Structure	Where it is	How to recognize it
Renal cortex	Outer layer, under the capsule	Grainy band around the edge
Renal medulla	Deep to the cortex	Made up of the pyramids
Renal pyramid	In the medulla	Striped cone with its base toward the cortex
Renal papilla	Tip of each pyramid	Points into a minor calyx
Renal column	Between neighboring pyramids	Cortex tissue reaching inward
Parenchyma	Cortex and medulla together	All the tissue that holds nephrons
Minor calyx	Around each papilla	Small cup
Major calyx	Where minor calyces join	Larger branch leading to the pelvis
Renal pelvis	In the renal sinus, at the hilum	Funnel that becomes the ureter
Renal sinus	Space just inside the hilum	Holds the pelvis, calyces, vessels, and fat

- Memory tip: renal columns are cortex. If a strip of tissue sits between two pyramids and looks like the outer layer, it is a renal column, even though it lies deep in the kidney.
- Sorting the parts: the cortex, renal columns, renal pyramids, and renal papillae are parenchyma. The minor calyces, major calyces, and renal pelvis are the spaces that collect urine. The papilla is the tip of a pyramid, so it belongs with the parenchyma, not with the urine spaces.
- The pyramids point their bases out toward the cortex and their tips in toward the calyces. An upside down pyramid is the usual slip on a drawing.

Competency 135. Nephron structure

In your course notes: Module 5 packet, Chapter 1, competency 135 (page 7).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Identify the renal corpuscle, glomerulus, glomerular capsule, podocytes, PCT, nephron loop, DCT, and collecting duct of a nephron.

The nephron is the microscopic unit that makes urine. Each kidney has roughly a million of them. A nephron has two main parts. The renal corpuscle filters the blood, and the renal tubule turns that filtrate into urine.

The renal corpuscle

The renal corpuscle is the filtering head of the nephron, and it always sits in the cortex. It is made of a ball of capillaries and the cup that surrounds it.

Glomerulus

The ball of capillaries. Its capillary walls are fenestrated (full of small pores), which makes them very leaky.

Glomerular capsule (Bowman's capsule)

The double-walled cup around the glomerulus. Its outer parietal layer is simple squamous epithelium and forms the outside wall of the cup. Its inner visceral layer hugs the glomerular capillaries and is made of podocytes.

Podocytes

The cells of the visceral layer. They have foot-like extensions (foot processes) that wrap around the capillaries. The narrow gaps between the feet are filtration slits. The slits are gaps between the podocyte feet, not holes in the capillary wall.

Capsular space

The space between the parietal and visceral layers. Filtrate collects in this space and then flows into the tubule.

The renal tubule and collecting duct

The renal tubule leaves the corpuscle as one long tube with three parts, and then empties into a collecting duct

that is shared with other nephrons.

Proximal convoluted tubule (PCT)

Comes first. It is coiled, stays in the cortex, and is lined with cuboidal cells that carry a brush border of microvilli. The brush border belongs on the proximal tubule, the first coiled segment, not the distal one.

Nephron loop (loop of Henle)

A U-shaped section that dips toward or into the medulla. It has a descending limb going down and an ascending limb coming back up.

Distal convoluted tubule (DCT)

The loop returns to the cortex and becomes the DCT, a second coiled section with cuboidal cells and few microvilli.

Collecting duct

The DCT empties into a collecting duct. Many nephrons drain into the same collecting duct, so it is shared among them. The collecting ducts run down through the pyramids and merge into larger papillary ducts, which open on the renal papilla.

Structure	Where it is	Key feature
Glomerulus	Renal corpuscle, cortex	Ball of fenestrated capillaries where blood is filtered
Glomerular capsule	Renal corpuscle, cortex	Double-walled cup with a parietal layer and a visceral layer
Podocytes	Visceral layer of the capsule	Foot-like extensions wrap the capillaries and leave filtration slits
Capsular space	Between the two capsule layers	Collects the filtrate
Proximal convoluted tubule	Cortex	First coiled section; cuboidal cells with a brush border
Nephron loop	Dips into the medulla	U-shaped, with a descending limb and an ascending limb
Distal convoluted tubule	Cortex	Second coiled section; few microvilli
Collecting duct	Cortex down through the pyramids	Shared by many nephrons; leads to papillary ducts

The path of the filtrate, in order

1. Glomerulus: blood is filtered out of the capillaries.
2. Capsular space: the filtrate collects inside the glomerular capsule.
3. Proximal convoluted tubule: the first coiled section of the tubule.
4. Descending limb of the nephron loop: the filtrate moves down toward the medulla.
5. Ascending limb of the nephron loop: the filtrate moves back up toward the cortex.
6. Distal convoluted tubule: the second coiled section of the tubule.
7. Collecting duct: fluid from many nephrons joins here.
8. Papillary duct: opens on the renal papilla and drains into a minor calyx.

- Memory tip: the segments are named in the order the filtrate meets them. Proximal means near the corpuscle, distal means farther away. Learn them in that order and you also know the path of the filtrate.

Competency 136. Cortical vs juxtamedullary nephrons

In your course notes: Module 5 packet, Chapter 1, competency 136 (page 8).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Compare cortical and juxtamedullary nephrons by loop depth and associated capillary beds.

Every nephron has the same parts. Nephrons differ in where the renal corpuscle sits and how deep the nephron loop reaches. The depth of the loop decides which capillary bed runs alongside the tubule.

Cortical nephrons

About 85 percent of all nephrons. Their renal corpuscles sit in the outer part of the cortex. Their loops are short and barely dip into the outer medulla. They are wrapped by the peritubular capillaries, a mesh of capillaries that surrounds the tubules in the cortex.

Juxtamedullary nephrons

About 15 percent. Juxta means next to, and their renal corpuscles sit in the deepest part of the cortex, right next to the medulla. Their loops are long and reach deep into the medulla, close to the renal papilla. Long, straight capillaries called the vasa recta run down and back up beside these loops. Recta means straight.

Both capillary beds are supplied by efferent arterioles leaving the glomeruli. The long loops of the juxtamedullary nephrons, together with the vasa recta, are what allow the kidney to make concentrated urine.

Feature	Cortical nephrons	Juxtamedullary nephrons
Share of all nephrons	About 85 percent	About 15 percent
Where the renal corpuscle sits	Outer cortex	Deep cortex, next to the medulla
Nephron loop	Short; barely enters the medulla	Long; reaches deep into the medulla
Capillary bed along the tubule	Peritubular capillaries	Vasa recta

- Memory tip: long loop, long straight vessels. The juxtamedullary nephron reaches deep, so its capillaries (the vasa recta) have to reach deep with it.
- Both kinds of renal corpuscle sit in the cortex. The difference is the outer cortex for the cortical nephron and the deep cortex next to the medulla for the juxtamedullary nephron.
- The juxtamedullary nephrons are the rare ones, about 15 percent, but they are the ones that concentrate the urine. Do not swap the percents, and do not swap the capillary beds.

Competency 137. JGA and filtration membrane

In your course notes: Module 5 packet, Chapter 1, competency 137 (page 9).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Identify the juxtaglomerular apparatus, macula densa, and JG cells, and name the three layers of the filtration membrane and what each holds back.

The juxtaglomerular apparatus

After the nephron loop climbs back to the cortex, the tubule passes right beside its own renal corpuscle, between the afferent and efferent arterioles. The spot where the tubule and the arterioles touch is the juxtaglomerular apparatus (JGA). It checks the fluid in the tubule and the blood pressure in the arteriole, and it helps adjust filtration.

Macula densa

A patch of tall, closely packed cells in the tubule wall, where the ascending limb of the loop ends and the distal convoluted tubule begins. Macula densa means dense spot, which describes how the crowded cells look on a slide. These cells sense the amount of sodium chloride in the fluid passing by.

Juxtaglomerular cells (JG cells, also called granular cells)

Modified smooth muscle cells, mostly in the wall of the afferent arteriole. Their granules hold the enzyme renin, which they release when blood pressure drops.

Structure	Where it is	What it does
Juxtaglomerular apparatus	Where the tubule touches the afferent and efferent arterioles of its own corpuscle	Monitors the tubule fluid and the arteriole and helps adjust filtration
Macula densa	Tubule wall, where the ascending limb meets the distal convoluted tubule	Senses the sodium chloride level of the fluid
Juxtaglomerular (granular) cells	Wall of the afferent arteriole	Release renin when blood pressure is low

- The juxtaglomerular cells sit in the afferent arteriole, the vessel going in, and the macula densa sits in the distal tubule. These two get swapped a lot.

The filtration membrane

Blood in the glomerulus is filtered into the capsular space across the filtration membrane. Fluid crosses three layers in a set order, and each layer holds back a different size of particle.

Layer (from the blood side to the capsular space)	What it is	What it holds back
Fenestrated endothelium	The pored wall of the glomerular capillary; lets plasma through	Blood cells and platelets
Basement membrane	The layer between the capillary wall and the podocytes	Large plasma proteins
Filtration slits of the podocytes	The gaps between podocyte feet, bridged by thin slit membranes	Most of the smaller (medium sized) proteins that get past the basement membrane

- What passes all three layers is the filtrate: water and small dissolved substances such as ions, glucose, and wastes like urea. Healthy filtrate has almost no protein and no blood cells.
- Protein or blood in the urine is a sign that the filtration membrane is damaged.
- Memory tip: each layer catches something smaller than the one before it: cells first, then big proteins, then the smaller proteins. Run the layers from the blood side out.

Competency 138. Kidney blood supply pathway

In your course notes: Module 5 packet, Chapter 1, competency 138 (page 10).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Trace blood through the kidney from the renal artery to the renal vein, naming each vessel in order.

The kidneys receive about a quarter of the blood the heart pumps at rest. Blood follows a fixed chain of arteries inward, passes through two capillary beds, and leaves through a chain of veins that runs back along the same route.

The names of the vessels tell you where they run. A pyramid with its surrounding cortex is a lobe, so interlobar vessels run between lobes, up through the renal columns. Arcuate means arched, and these vessels arch over the bases of the pyramids at the border between the cortex and medulla. Cortical radiate vessels radiate outward into the cortex. Some books call them interlobular vessels.

Renal artery to renal vein, in order

1. Renal artery: branches off the abdominal aorta and enters the kidney at the hilum.
2. Segmental arteries: branches of the renal artery inside the renal sinus, before the blood reaches the tissue.
3. Interlobar arteries: run up through the renal columns, between the pyramids.
4. Arcuate arteries: arch over the bases of the pyramids, along the border between cortex and medulla.

5. Cortical radiate arteries: also called interlobular arteries; radiate outward into the cortex.
6. Afferent arteriole: carries blood into one glomerulus.
7. Glomerulus: the first capillary bed, where blood is filtered.
8. Efferent arteriole: carries blood out of the glomerulus.
9. Peritubular capillaries and vasa recta: the second capillary bed. Peritubular capillaries wrap the tubules in the cortex, and the vasa recta run beside the long loops in the medulla.
10. Cortical radiate veins: drain the cortex and run back toward the medulla.
11. Arcuate veins: arch over the bases of the pyramids.
12. Interlobar veins: run down through the renal columns toward the renal sinus.
13. Renal vein: leaves the kidney at the hilum and empties into the inferior vena cava.

- This path is unusual in one way. In most of the body, a capillary bed drains into a venule. The glomerulus has an arteriole on both sides, the afferent arteriole going in and the efferent arteriole coming out. Blood then passes through a second capillary bed before it reaches any vein.
- Memory tip: blood meets two capillary beds in the kidney, one after the other: the glomerulus, where filtration happens, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them, not the afferent arteriole.
- On the way out, the veins repeat the artery names in reverse: cortical radiate, arcuate, interlobar, renal.
- Only the interlobar arteries run between the pyramids. The segmental arteries branch earlier, in the renal sinus.

Competency 139. Ureters, bladder, urethra and urine path

In your course notes: Module 5 packet, Chapter 1, competency 139 (page 11).

Dr. Rennie's video for this part: Urinary system concept video.

You should be able to: Trace urine from the collecting duct to the outside and identify the ureter, bladder wall layers, trigone, detrusor, sphincters, and male and female urethra.

Once fluid leaves the collecting duct it is urine, and nothing along the rest of the path changes it much. The rest of the system collects the urine, carries it, stores it, and releases it.

The path of urine, in order

1. Collecting duct: gathers fluid from many nephrons and carries it through the pyramid.
2. Papillary duct: opens on the renal papilla.
3. Minor calyx: cup that catches urine from one papilla.
4. Major calyx: forms where several minor calyces join.
5. Renal pelvis: funnel that collects urine from the major calyces and narrows into the ureter.
6. Ureter: carries urine down to the bladder.
7. Urinary bladder: stores the urine.
8. Urethra: carries urine out of the body.
9. External urethral orifice: the opening to the outside.

The ureters

- Each ureter is a muscular tube about 30 cm long. It leaves the renal pelvis at the hilum, runs down the posterior abdominal wall behind the peritoneum, and enters the posterior wall of the bladder.
- The ureter passes through the bladder wall at a slant. When the bladder fills, pressure squeezes this slanted part closed, which keeps urine from flowing back up toward the kidney.
- The ureter wall has three layers. The inner mucosa is lined with transitional epithelium (urothelium), which can stretch. The middle layer is smooth muscle, which pushes urine along by peristalsis. The outer layer is a fibrous adventitia.

The urinary bladder

The urinary bladder is a hollow, muscular sac in the pelvic cavity, just behind the pubic symphysis. In males it sits in front of the rectum and above the prostate. In females it sits in front of the vagina and below the uterus. Only its upper surface is covered by peritoneum.

Layer (from the inside out)	What it is	Key feature
Mucosa	Transitional epithelium (urothelium) on a layer of connective tissue (lamina propria)	Stretches as the bladder fills; folds into rugae when the bladder is empty
Detrusor muscle	The muscular layer, made of layers of smooth muscle	Contracts to squeeze urine out
Adventitia	Outer fibrous connective tissue	Over the upper surface, the peritoneum (a serosa) covers the bladder

Trigone

A smooth triangle on the floor of the bladder. Its three corners are the two ureteral openings at the back (the upper corners) and the internal urethral orifice at the front (the lowest point), where the urethra begins. The trigone has no rugae, so it stays smooth even when the bladder is empty.

Internal urethral sphincter

A thickening of smooth muscle at the neck of the bladder, where the urethra begins. It is involuntary, so you cannot control it.

External urethral sphincter

Skeletal muscle that surrounds the urethra as it passes through the pelvic floor. It is voluntary and gives conscious control over urination.

- The internal sphincter is smooth muscle you cannot control; the external one is skeletal muscle you can. These two get swapped, so check them.

The urethra

The urethra carries urine from the bladder to the outside. It is very different in females and males, and the male urethra is the longer one.

Feature	Female urethra	Male urethra
Length	About 4 cm	About 20 cm
What it carries	Urine only	Urine and semen
Parts	One short tube	Prostatic urethra through the prostate, membranous urethra through the deep muscles of the perineum, and spongy urethra through the corpus spongiosum of the penis, in that order from the bladder down
External urethral orifice	Between the clitoris and the vaginal opening	At the tip of the glans penis

Competency 140. Clinical anatomy of urinary disorders

In your course notes: Module 5 packet, Chapter 1, competency 140 (page 12).

You should be able to: Relate urinary disorders such as nephroptosis, kidney stones, and UTI to the anatomical structure or pathway each one affects.

Urinary disorders make the most sense when you place each one on the structures you already know. Some damage the filtering parts of the nephron. Others block, infect, or weaken the tubes that carry and hold urine.

Any blockage along the path makes urine back up into the parts above it. When urine backs up into the renal pelvis and calyces, they swell. This swelling is called hydronephrosis.

Disorder	Structure or pathway affected	What changes
Nephroptosis	Adipose capsule and renal fascia; the ureter below	The kidney drops out of its normal position, often when its cushion of fat is thin or lost. The ureter can kink and block urine flow
Kidney stones (renal calculi)	Form in the calyces and renal pelvis; lodge in the ureter	Minerals harden into stones. A stone tends to stick at the narrow points of the ureter: where it leaves the renal pelvis, where it crosses the iliac vessels at the pelvic brim, and where it enters the bladder
Urinary tract infection (UTI)	Urethra, bladder, and, if it climbs, the ureter and kidney	Bacteria, most often from the intestine (such as E. coli), climb up the urine path and inflame its lining. Infection of the bladder is cystitis. Infection that reaches the renal pelvis and kidney tissue is pyelonephritis
Glomerulonephritis	Glomeruli	The glomeruli become inflamed, which damages the filtration membrane. Protein and blood can leak into the urine
Renal failure	Nephrons of both kidneys	The kidneys lose their ability to filter the blood, either suddenly (acute) or slowly over time (chronic)
Polycystic kidney disease	Renal parenchyma (the tubules)	An inherited disorder in which many fluid-filled cysts form and enlarge the kidneys, crowding out working tissue
Urinary incontinence	Detrusor muscle, urethral sphincters, and pelvic floor	Loss of voluntary control over urination. In stress incontinence a weak pelvic floor and external sphincter leak with coughing or lifting. In urge incontinence the detrusor contracts too readily. In overflow incontinence the bladder cannot empty fully, so it overfills and leaks. In functional incontinence the urinary organs work, but the person cannot reach a toilet in time

What this means for patients

- Position and pathway predict the problem. The kidney depends on its adipose capsule to stay in place. When that fat is lost, the kidney loses its support and can drop, and the ureter can kink and block the flow of urine.
- Urine travels one narrow path. A kidney stone often causes little pain while it sits in the renal pelvis. Pain usually starts when the stone moves into the ureter, where it scrapes and blocks a tube only a few millimeters wide. Where the stone sits along that path shapes where the pain is felt.
- The two urethras differ, and so does the risk. The female urethra is short and opens close to the anus, which gives bacteria a shorter climb to the bladder. This is one reason urinary tract infections are more common in women.
- A urinary tract infection can sit in the urethra, the bladder, or the kidney. It is not tied to only one spot.

Worked questions

Nine questions in the style of the lecture exam, at least one for each competency. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 133

Which layer around the kidney is the outermost, anchoring the kidney and adrenal gland to the posterior abdominal wall?

A. Renal capsule

B. Adipose capsule

C. Renal fascia

D. Parietal peritoneum

Answer: C. From the kidney outward the layers go capsule, fat, fascia. The renal fascia is a sheet of dense connective tissue farthest out, and it ties the kidney and the adrenal gland to the body wall.

Why the others are wrong

- **A** The renal capsule is the innermost layer, the thin, tough fibrous covering right on the kidney surface. It holds the kidney's shape.
- **B** The adipose capsule is the middle layer of fat (perirenal fat). It cushions the kidney and helps hold it in place, but the fascia is still outside it.
- **D** The parietal peritoneum lies in front of the kidney, not around it. The kidney is retroperitoneal, meaning it sits behind the peritoneum.

QUESTION 2 · COMPETENCY 134

A strip of grainy tissue lies deep in the kidney between two renal pyramids. What is it?

A. Renal column

B. Renal papilla

C. Minor calyx

D. Renal pelvis

Answer: A. Renal columns are cortex tissue reaching inward between neighboring pyramids. They look like the outer layer because they are the outer layer, extended deep into the kidney.

Why the others are wrong

- **B** The renal papilla is the narrow tip of a pyramid, pointing into a minor calyx. It is part of the pyramid, not a strip between pyramids.
- **C** A minor calyx is a small cup of collecting space around a papilla. It is a space for urine, not grainy tissue.
- **D** The renal pelvis is the funnel in the renal sinus that becomes the ureter. It collects urine from the major calyces and is not between the pyramids.

QUESTION 3 · COMPETENCY 135

Which part of the nephron is the first coiled segment and carries a brush border of microvilli?

A. Distal convoluted tubule

B. Collecting duct

C. Nephron loop

D. Proximal convoluted tubule

Answer: D. The proximal convoluted tubule leaves the glomerular capsule first. It is coiled, stays in the cortex, and its cuboidal cells carry a brush border of microvilli.

Why the others are wrong

- **A** The distal convoluted tubule is the second coiled segment, after the loop. Its cuboidal cells have few microvilli.
- **B** The collecting duct comes after the tubule and is shared by many nephrons. It runs down through the pyramid; it is not coiled.
- **C** The nephron loop is the U-shaped section with a descending and an ascending limb. It dips toward the medulla and is not coiled.

QUESTION 4 · COMPETENCY 136

Which capillary bed runs beside the long nephron loops of the juxtamedullary nephrons?

A. Glomerulus

B. Vasa recta

C. Peritubular capillaries

D. Arcuate arteries

Answer: B. The vasa recta are long, straight capillaries that run down and back up beside the long loops in the medulla. Long loop, long straight vessels. Together they let the kidney make concentrated urine.

Why the others are wrong

- **A** The glomerulus is the first capillary bed, inside the renal corpuscle in the cortex. Every nephron has one; it does not run beside the loop.
- **C** The peritubular capillaries are the mesh around the tubules in the cortex, and they go with the cortical nephrons, whose short loops barely enter the medulla.
- **D** Arcuate arteries are arteries, not capillaries. They arch over the bases of the pyramids at the border between cortex and medulla.

QUESTION 5 · COMPETENCY 137

Where are the juxtaglomerular (granular) cells, and what do they release?

A. In the wall of the afferent arteriole; renin

B. In the wall of the distal convoluted tubule; renin

C. In the wall of the afferent arteriole; sodium chloride

D. In the wall of the efferent arteriole; urea

Answer: A. JG cells are modified smooth muscle cells, mostly in the wall of the afferent arteriole, the vessel going into the glomerulus. Their granules hold renin, which they release when blood pressure drops.

Why the others are wrong

- **B** The distal convoluted tubule wall holds the macula densa, the dense spot of tall cells that senses sodium chloride. It does not release renin.
- **C** The location is right, but JG cells release renin. Sodium chloride is what the macula densa senses in the tubule fluid, not something a cell releases.
- **D** The JG cells sit in the afferent arteriole, not the efferent, and urea is a waste in the filtrate, not something the JGA releases.

QUESTION 6 · COMPETENCY 137

Which layer of the filtration membrane holds back large plasma proteins?

A. Fenestrated endothelium

B. Filtration slits of the podocytes

C. Macula densa

D. Basement membrane

Answer: D. The basement membrane is the middle layer, between the capillary wall and the podocytes. It stops the large plasma proteins. Each layer catches something smaller than the one before: cells, then large proteins, then medium sized proteins.

Why the others are wrong

- **A** The fenestrated endothelium is the first layer on the blood side. Its pores let plasma through but hold back blood cells and platelets.
- **B** The filtration slits, bridged by slit membranes between the podocyte feet, are the last layer. They hold back most of the smaller proteins that get past the basement membrane.
- **C** The macula densa is not part of the filtration membrane at all. It is the sensing patch in the distal tubule wall, part of the juxtaglomerular apparatus.

QUESTION 7 · COMPETENCY 138

In the kidney, which vessel carries blood from the first capillary bed to the second?

A. Afferent arteriole

B. Cortical radiate artery

C. Efferent arteriole

D. Arcuate vein

Answer: C. The glomerulus is the first capillary bed and the peritubular capillaries and vasa recta are the second. The efferent arteriole carries blood out of the glomerulus and into that second bed. It is the one vessel between them.

Why the others are wrong

- **A** The afferent arteriole carries blood into the glomerulus. It comes before the first capillary bed, not between the two.
- **B** The cortical radiate (interlobular) artery radiates out into the cortex and gives rise to the afferent arterioles. It is upstream of both capillary beds.
- **D** The arcuate vein drains the cortical radiate veins and arches over the pyramid bases. It comes after the second capillary bed, on the way out.

QUESTION 8 · COMPETENCY 139

Which statement about the urethral sphincters is correct?

A. The internal sphincter is skeletal muscle and voluntary

B. The internal sphincter is smooth muscle and involuntary

C. The external sphincter is smooth muscle and involuntary

D. Both sphincters are smooth muscle

Answer: B. The internal urethral sphincter is a thickening of smooth muscle at the neck of the bladder, and it works without conscious control. The external urethral sphincter is skeletal muscle at the pelvic floor and is the one you control.

Why the others are wrong

- **A** That describes the external sphincter, not the internal one. The internal sphincter is smooth muscle you cannot control.
- **C** The external sphincter is skeletal muscle that surrounds the urethra as it passes through the pelvic floor. It is voluntary and gives conscious control over urination.
- **D** Only the internal sphincter is smooth muscle. The external one is skeletal muscle, which is why it can be controlled.

QUESTION 9 · COMPETENCY 140

A patient's kidney has dropped below its normal position and the ureter has kinked. Which structure has most likely been lost or thinned?

A. Renal capsule

B. Renal cortex

C. Glomeruli

D. Adipose capsule

Answer: D. This is nephroptosis. The kidney depends on its adipose capsule, the fat layer, to stay in place. When that fat is thin or lost, the kidney loses its support and drops, the ureter can kink, and urine flow is blocked.

Why the others are wrong

- **A** The renal capsule is the fibrous covering on the kidney surface. It holds the kidney's shape; it is not the cushion that holds the kidney in position.
- **B** The renal cortex is working tissue inside the kidney, where the renal corpuscles and coiled tubules sit. Losing it would affect filtering, not position.
- **C** The glomeruli are the filtering capillary balls. Glomerulonephritis inflames them and lets protein and blood leak into the urine, but it does not move the kidney.